

H4 Senior Extremum — 25% Control Zone Approach Pattern

1. Purpose

The purpose of this pattern is to identify significant H4 ZigZag extremums, build a **25% Control Zone Level** relative to each confirmed extremum, and detect when the market subsequently approaches this level.

The pattern consists of three main stages:

1. Identify a **Senior H4 Extremum**.
 2. Build the **25% Control Zone Level** from this extremum.
 3. Detect and record a future **25% Level Approach Event**.
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2. Dependencies

The pattern uses the existing logic defined for:

- H4 ZigZag extremums;
- Forward Margin Zone distance (**FMZ**);
- Initial Margin Zone distance (**IMZ**);
- Control / Margin Zone construction.

For this pattern, only the **nearest boundary of the Control Zone** is required.

Because the nearest boundary of the Control Zone is located at the **FMZ** distance from the ZigZag extremum:

Distance to nearest Control Zone boundary = FMZ

3. Senior H4 Extremum Identification

3.1 General Rule

Use the H4 ZigZag to identify local extremums.

A ZigZag extremum is classified as a **Senior Extremum** when the nearest extremum of the same type on each side of it is weaker than the candidate extremum.

In other words, the Senior Extremum must dominate one same-type ZigZag extremum to its left and one same-type ZigZag extremum to its right.

3.2 Senior Maximum

Let:

- H_0 = candidate ZigZag local maximum;
- H_{-1} = nearest previous ZigZag local maximum;
- H_{+1} = nearest subsequent ZigZag local maximum.

H_0 is a **Senior Maximum** if:

$$H_0 > H_{-1}$$

and

$$H_0 > H_{+1}$$

Therefore:

$$H_{-1} < H_0 > H_{+1}$$

Both neighbouring local maximums must be below the Senior Maximum.

3.3 Senior Minimum

Let:

- L_0 = candidate ZigZag local minimum;
- L_{-1} = nearest previous ZigZag local minimum;
- L_{+1} = nearest subsequent ZigZag local minimum.

L_0 is a **Senior Minimum** if:

$$L_0 < L_{-1}$$

and

$$L_0 < L_{+1}$$

Therefore:

$$L_{-1} > L_0 < L_{+1}$$

Both neighbouring local minimums must be above the Senior Minimum.

3.4 Confirmation Moment

A Senior Extremum cannot be considered confirmed until the required same-type extremum to its right has been formed by the ZigZag.

For example:

H-1 → H0 → H+1

H0 becomes a confirmed Senior Maximum only after H+1 has been confirmed.

The algorithm must therefore store separately:

- ExtremumTime — time when H0 or L0 originally occurred;
- ConfirmationTime — time when the right-side extremum required to confirm the pattern became available.

No-lookahead rule

Any subsequent pattern statistics must use only information that was available after ConfirmationTime.

The backtest must not treat the Senior Extremum as known at ExtremumTime.

4. Building the 25% Control Zone Level

4.1 Concept

The **25% Control Zone Level** is not located inside the Control Zone.

It is a horizontal price level located **25% of the distance from the Senior Extremum toward the nearest boundary of the Control Zone**.

The nearest boundary of the full Control Zone is located at the FMZ distance from the extremum.

Therefore:

$$\text{Distance25} = \text{FMZ} \times 0.25$$

4.2 Senior Maximum

For a Senior Maximum, the Control Zone is located below the extremum.

Let:

- E = Senior Maximum price;
- FMZ = distance from the extremum to the nearest Control Zone boundary.

Then:

$$Level_{25} = E - (FMZ \times 0.25)$$

4.3 Senior Minimum

For a Senior Minimum, the Control Zone is located above the extremum.

Then:

$$Level_{25} = E + (FMZ \times 0.25)$$

5. 25% Level Approach Event

5.1 Objective

After the Senior Extremum and its corresponding 25% Level have been confirmed, monitor future market prices.

An event must be generated when the market approaches the 25% Level within the defined tolerance.

5.2 Approach Tolerance

The proximity threshold is defined relative to the distance between the Senior Extremum and the 25% Level.

Let:

- E = Senior Extremum price;
- $Level_{25}$ = calculated 25% Level;
- $Distance_{25}$ = distance between the extremum and the 25% Level.

Calculate:

$$Distance_{25} = ABS(E - Level_{25})$$

Because:

$$Distance_{25} = FMZ \times 0.25$$

The **Approach Tolerance** is:

$$\text{Tolerance} = \text{Distance25} \times 0.10$$

Therefore:

$$\text{Tolerance} = \text{FMZ} \times 0.025$$

The tolerance represents **10% of the distance between the Senior Extremum and the 25% Level**.

5.3 Approach Range

The Approach Range is constructed symmetrically around the 25% Level:

$$\text{LowerBound} = \text{Level25} - \text{Tolerance}$$

$$\text{UpperBound} = \text{Level25} + \text{Tolerance}$$

Therefore, an Approach Event occurs whenever the market enters:

$$[\text{LowerBound} ; \text{UpperBound}]$$

Equivalent condition:

$$\text{ABS}(\text{MarketPrice} - \text{Level25}) \leq \text{Distance25} \times 0.10$$

Important

The 10% threshold must **not** be calculated as a percentage of the absolute market price.

For example, it is incorrect to use:

$$\text{Tolerance} = \text{Level25} \times 0.10$$

The tolerance must always be derived from the geometry of the pattern itself:

$$\text{Tolerance} = \text{ABS}(\text{ExtremumPrice} - \text{Level25}) \times 0.10$$

6. Market Price Detection

The algorithm must check whether the traded market range enters the Approach Range.

For OHLC candle data, an event occurs if the candle price range intersects the Approach Range.

Condition:

```
CandleHigh >= LowerBound  
AND  
CandleLow <= UpperBound
```

This allows the pattern to detect an approach even if the market entered the threshold intraday but did not close inside it.

The event must therefore not depend only on the candle Close price.

7. Event Timing

Monitoring begins only after:

```
SeniorExtremum.ConfirmationTime
```

The algorithm scans subsequent market data chronologically.

When the Approach Range is reached for the first time, create:

25% Control Zone Approach Event

The event timestamp should correspond to the first candle that entered the Approach Range.

8. Day Identification

Each detected event must be mapped to the corresponding trading day.

Store at minimum:

- Instrument;
- Senior Extremum type: `HIGH` / `LOW` ;
- Senior Extremum price;
- Senior Extremum timestamp;
- Senior Extremum confirmation timestamp;
- FMZ;
- IMZ;
- 25% Level price;
- Distance between Extremum and 25% Level;
- Approach Tolerance;
- Approach Range lower boundary;
- Approach Range upper boundary;

- First approach timestamp;
 - Trading date of the approach;
 - Market price / candle range that triggered the event.
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9. Chart Visualization

For every confirmed Senior Extremum:

Senior Extremum

Mark the H4 ZigZag extremum used as the pattern anchor.

25% Level

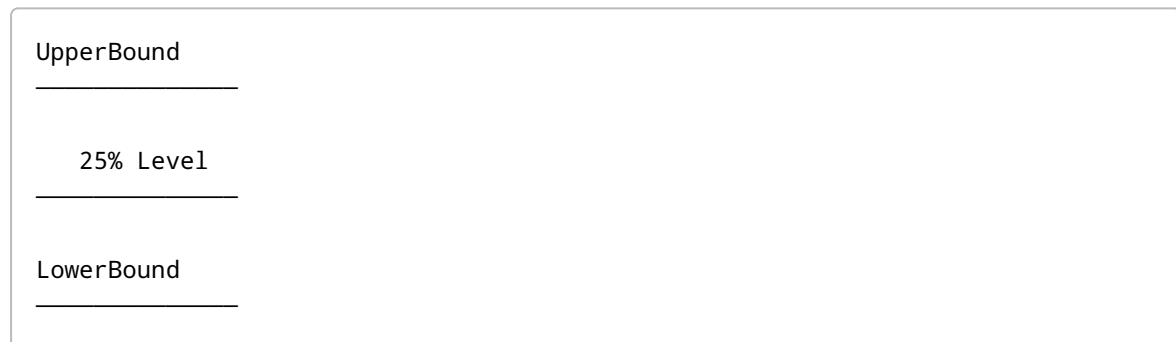
Draw a horizontal line at:

Level25

The line should extend forward in time starting from the Senior Extremum.

Approach Range

Optionally visualize the tolerance corridor around the 25% Level:



The distance from the 25% Level to either boundary equals:

$10\% \times \text{Distance25}$

Approach Event

When the market first enters the Approach Range:

- mark the corresponding candle;
 - highlight the trading day on the chart;
 - associate the event with its originating Senior Extremum and 25% Level.
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10. Processing Logic

For each instrument:

1. Calculate the H4 ZigZag.
2. Extract ZigZag local maximums and minimums.
3. For every candidate extremum, locate:
 4. the nearest same-type ZigZag extremum to the left;
 5. the nearest same-type ZigZag extremum to the right.
6. Check the Senior Extremum condition.

For maximum:

```
PreviousHigh < CandidateHigh  
AND  
NextHigh < CandidateHigh
```

For minimum:

```
PreviousLow > CandidateLow  
AND  
NextLow > CandidateLow
```

1. Once the right-side extremum exists, mark the candidate as a confirmed Senior Extremum.
2. Calculate the `FMZ` applicable to the extremum.
3. Calculate:

```
Distance25 = FMZ × 0.25
```

1. Build the 25% Level.

For maximum:

```
Level25 = ExtremumPrice - Distance25
```

For minimum:

$\text{Level25} = \text{ExtremumPrice} + \text{Distance25}$

1. Calculate the Approach Tolerance:

$\text{Tolerance} = \text{Distance25} \times 0.10$

1. Calculate the Approach Range:

$\text{LowerBound} = \text{Level25} - \text{Tolerance}$

$\text{UpperBound} = \text{Level25} + \text{Tolerance}$

1. Starting from `ConfirmationTime`, scan subsequent market data.
2. Detect the first candle whose price range intersects the Approach Range.
3. Record a **25% Control Zone Approach Event**.
4. Highlight the trading day containing that event on the chart.

11. Validation Rules

Senior Maximum

A candidate maximum is valid only when both neighbouring same-type ZigZag maximums are lower.

Senior Minimum

A candidate minimum is valid only when both neighbouring same-type ZigZag minimums are higher.

No Lookahead

The Senior Extremum may only become active after its right-side confirming extremum has appeared.

25% Level

The 25% Level must always lie between:

- the Senior Extremum;
- the nearest edge of the full Control Zone.

It must not be calculated as 25% of the width between FMZ and IMZ.

Direction

For a Senior Maximum:

$$\text{Level25} < \text{ExtremumPrice}$$

For a Senior Minimum:

$$\text{Level25} > \text{ExtremumPrice}$$

Approach Tolerance

The approach tolerance must equal:

$$10\% \times \text{ABS}(\text{ExtremumPrice} - \text{Level25})$$

or equivalently:

$$2.5\% \times \text{FMZ}$$

It must not be calculated as a percentage of the absolute price of the instrument.

Event

An Approach Event must be generated only from market data occurring after the Senior Extremum has been confirmed.

Event Day

The chart must clearly identify the trading day on which the first valid approach occurred.